EE 5239 Nonlinear Optimization: Stochastic Gradient Descent (b)

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Vanilla SGD Algorithm: Convergence Analysis

$$\begin{array}{ll} \text{minimize}_{\boldsymbol{x}} & F(\boldsymbol{x}) := \mathbb{E}[f(\boldsymbol{x}, \xi)] \\ \text{subject to} & \boldsymbol{x} \in \mathbb{R}^n \end{array}$$

- $f(\boldsymbol{x}, \xi)$ (objective or cost function) is differentiable for every ξ
- ξ is a random variables.

Example

Consider the finite-sum problem we discussed before

$$
\begin{aligned}
\text{minimize}_{\boldsymbol{x}} \quad & F(\boldsymbol{x}) := \frac{1}{N} \sum_{i=1}^{N} f(\boldsymbol{x}, \{\mathbf{a}_i, b_i\}) \\
\text{subject to} \quad & \boldsymbol{x} \in \mathbb{R}^n
\end{aligned}
$$

- where $\{\mathbf{a}_i, b_i\}_{i=1}^{N}$ are N random samples
- If we draw the indices i from a uniform distribution over $i = 1 \cdots N$, then

$$
F(\boldsymbol{x}) = \mathbb{E}_i[f(\boldsymbol{x}, \{\mathbf{a}_i, b_i\})].
$$

The SGD Algorithm

- Consider the following GD algorithm

$$\begin{aligned} \boldsymbol{x}^{r+1} &= \boldsymbol{x}^r - \nabla F(\boldsymbol{x}^r) \\ &= \boldsymbol{x}^r - \nabla \mathbb{E}[f(\boldsymbol{x}^r, \xi)] \\ &= \boldsymbol{x}^r - \mathbb{E}[\nabla f(\boldsymbol{x}^r, \xi)] \end{aligned}$$

- The gradient cannot be exactly evaluated
- Use a sample to approximate it

The SGD Algorithm

- The SGD algorithm, first sample $\mathbf{g}^r$ as an unbiased estimator of $\mathbb{E}[\nabla f(\boldsymbol{x}^r, \xi)]$
- Then perform

$$\boldsymbol{x}^{r+1} = \boldsymbol{x}^r - \eta^r \mathbf{g}^r.$$

Convergence Analysis (f is Strongly Convex)

- Let $F(\boldsymbol{x})$ be L-smooth, and
  - 1) $F(\boldsymbol{x})$ is strongly convex, with constant μ
  - 2) $F(\boldsymbol{x})$ is non-convex
- $F(\boldsymbol{x})$ is lower bounded, $F(\boldsymbol{x}) \geq \underline{F}$, $\forall\ \boldsymbol{x}$
- Let $\mathbf{g}^r$ be an unbiased estimator of $\nabla F(\boldsymbol{x}^r)$:

$$\mathbb{E}[\mathbf{g}^r] = \nabla F(\boldsymbol{x}^r). \tag{1.1}$$

- Assume

$$\mathbb{E}[\|\mathbf{g}^r\|^2] \leq \sigma_g^2 + c_g \|\nabla F(\boldsymbol{x}^r)\|^2 \tag{1.2}$$

Convergence Analysis

Theorem 1.1 (Convergence of SGD for Strongly Convex f)

Under the assumption in the previous page, where f is μ strongly convex, if $c_g = 0$, and $\eta_r \leq \frac{1}{\mu r}$, then SGD achieves the following rate

$$\mathbb{E}[\|\boldsymbol{x}^{T+1} - \boldsymbol{x}^*\|^2] = \mathcal{O}(1/T); \tag{1.3}$$

Further, if $c_g \neq 0$, then if we choose $\eta_r \leq \frac{1}{c_g L^2 \mu r}$, we have

$$\mathbb{E}[\|\boldsymbol{x}^{T+1} - \boldsymbol{x}^*\|^2] = \mathcal{O}(1/T) \tag{1.4}$$

Proof Steps

- First, let us suppose that $c_g = 0$
- From the fact that F is strongly convex, we have

$$F(\boldsymbol{x}^*) - F(\boldsymbol{x}^r) \geq \langle \nabla F(\boldsymbol{x}^r), \boldsymbol{x}^* - \boldsymbol{x}^r \rangle + \frac{\mu}{2}\|\boldsymbol{x}^r - \boldsymbol{x}^*\|^2$$
$$F(\boldsymbol{x}^r) - F(\boldsymbol{x}^*) \geq \langle \nabla F(\boldsymbol{x}^*), \boldsymbol{x}^r - \boldsymbol{x}^* \rangle + \frac{\mu}{2}\|\boldsymbol{x}^r - \boldsymbol{x}^*\|^2$$

- Adding these together, we obtain

$$\begin{aligned} &\langle \nabla F(\boldsymbol{x}^r) - \nabla F(\boldsymbol{x}^*), \boldsymbol{x}^r - \boldsymbol{x}^* \rangle \\ &= \langle \nabla F(\boldsymbol{x}^r), \boldsymbol{x}^r - \boldsymbol{x}^* \rangle \geq \mu \|\boldsymbol{x}^r - \boldsymbol{x}^*\|^2 \end{aligned} \tag{1.5}$$

Proof Steps ($c_g = 0$)

- Similarly as the subgradient method analysis, we have

$$\begin{aligned} \|\boldsymbol{x}^* - \boldsymbol{x}^{r+1}\|^2 &= \|\boldsymbol{x}^* - \boldsymbol{x}^r\|^2 + 2\langle \boldsymbol{x}^* - \boldsymbol{x}^r, \boldsymbol{x}^r - \boldsymbol{x}^{r+1}\rangle + \|\boldsymbol{x}^r - \boldsymbol{x}^{r+1}\|^2 \\ &= \|\boldsymbol{x}^* - \boldsymbol{x}^r\|^2 + 2\eta\langle \boldsymbol{x}^* - \boldsymbol{x}^r, \mathbf{g}^r\rangle + \eta^2\|\mathbf{g}^r\|^2 \end{aligned}$$

- Taking an expectation, we obtain

$$\begin{aligned} \mathbb{E}[\|\boldsymbol{x}^* - \boldsymbol{x}^{r+1}\|^2] &= \mathbb{E}[\|\boldsymbol{x}^* - \boldsymbol{x}^r\|^2] + 2\eta\mathbb{E}[\langle \boldsymbol{x}^* - \boldsymbol{x}^r, \mathbf{g}^r\rangle] + \eta^2\mathbb{E}[\|\mathbf{g}^r\|^2] \\ &= \mathbb{E}[\|\boldsymbol{x}^* - \boldsymbol{x}^r\|^2] + 2\eta\langle \boldsymbol{x}^* - \boldsymbol{x}^r, \nabla F(\boldsymbol{x}^r)\rangle + \eta^2\mathbb{E}[\|\mathbf{g}^r\|^2] \\ &\leq (1 - 2\mu\eta)\mathbb{E}[\|\boldsymbol{x}^* - \boldsymbol{x}^r\|^2] + \eta^2\sigma_g^2 \end{aligned}$$

  where the last inequality comes from (1.5) and $c_g = 0$.

Proof Steps ($c_g = 0$)

- The final rate is proven by induction.
- At iteration $r = 1$, we have

$$\mathbb{E}\|\boldsymbol{x}^1 - \boldsymbol{x}^*\|^2 \leq \max\{\mathbb{E}\|\boldsymbol{x}^0 - \boldsymbol{x}^*\|^2, \sigma_g^2/\mu\} := L^0 \tag{1.6}$$

- Suppose for iteration r, the desired rate holds true, then for iteration $r+1$ (and use $\eta_r \leq \frac{1}{r\mu^2}$)

$$\begin{aligned}
\mathbb{E}[\|\boldsymbol{x}^{r+1} - \boldsymbol{x}^*\|^2] &\leq (1 - 2/r)\mathbb{E}[\|\boldsymbol{x}^r - \boldsymbol{x}^*\|^2] + 1/(\mu^2 r^2)\sigma_g^2 \\
&\leq (1 - 2/r)L^0/r + 1/(\mu^2 r^2)\sigma_g^2 \\
&\leq (1/r - 2/r^2)L^0 + L^0/r^2 \leq (1/r - 1/r^2)L^0 \\
&\leq \frac{L^0}{r+1}
\end{aligned}$$

Proof Steps ($c_g \neq 0$)

- Now consider the case $c_g \neq 0$
- Similarly as before, we have

$$\begin{aligned}\|\boldsymbol{x}^* - \boldsymbol{x}^{r+1}\|^2 &= \|\boldsymbol{x}^* - \boldsymbol{x}^r\|^2 + 2\langle \boldsymbol{x}^* - \boldsymbol{x}^r, \boldsymbol{x}^r - \boldsymbol{x}^{r+1}\rangle + \|\boldsymbol{x}^r - \boldsymbol{x}^{r+1}\|^2 \\ &= \|\boldsymbol{x}^* - \boldsymbol{x}^r\|^2 - 2\eta\langle \boldsymbol{x}^* - \boldsymbol{x}^r, \mathbf{g}^r\rangle + \eta^2\|\mathbf{g}^r\|^2\end{aligned}$$

- Taking an expectation, we obtain

$$\begin{aligned}\mathbb{E}[\|\boldsymbol{x}^* - \boldsymbol{x}^{r+1}\|^2] &= \mathbb{E}[\|\boldsymbol{x}^* - \boldsymbol{x}^r\|^2] - 2\eta\mathbb{E}[\langle \boldsymbol{x}^* - \boldsymbol{x}^r, \mathbf{g}^r\rangle] + \eta^2\mathbb{E}[\|\mathbf{g}^r\|^2] \\ &= \mathbb{E}[\|\boldsymbol{x}^* - \boldsymbol{x}^r\|^2] - 2\eta\mathbb{E}[\langle \boldsymbol{x}^* - \boldsymbol{x}^r, \nabla F(\boldsymbol{x}^r)\rangle] + \eta^2\mathbb{E}[\|\mathbf{g}^r\|^2] \\ &\overset{(1.5)}{\leq} (1 - 2\mu\eta)\mathbb{E}[\|\boldsymbol{x}^* - \boldsymbol{x}^r\|^2] + \eta^2\mathbb{E}[\|\mathbf{g}^r\|^2]\end{aligned}$$

where the second equality uses the conditional expectation $\mathbb{E}[g] = \mathbb{E}[\mathbb{E}[g|x^1, \cdots, x^r]]$.

Proof Steps ($c_g \neq 0$)

- Next let us bound $\mathbb{E}[\|\mathbf{g}^r\|^2]$ as (by using the ubiaseness)

$$\begin{aligned}\mathbb{E}[\|\mathbf{g}^r\|^2] &\le \sigma_g^2 + c_g\|\nabla F(\boldsymbol{x}^r)\|^2 \\ &= \sigma_g^2 + c_g\|\nabla F(\boldsymbol{x}^r) - \nabla F(\boldsymbol{x}^*)\|^2 \\ &\le \sigma_g^2 + c_g L^2\|\boldsymbol{x}^r - \boldsymbol{x}^*\|^2.\end{aligned}$$

- Combining the previous two results, we have

$$\begin{aligned}\mathbb{E}[\|\boldsymbol{x}^* - \boldsymbol{x}^{r+1}\|^2] &\le (1-2\mu\eta)\mathbb{E}[\|\boldsymbol{x}^* - \boldsymbol{x}^r\|^2] + \eta^2(\sigma_g^2 + c_g L^2\|\boldsymbol{x}^r - \boldsymbol{x}^*\|^2) \\ &= {\color{red}(1-2\mu\eta-\eta^2 c_g L^2)}\mathbb{E}[\|\boldsymbol{x}^* - \boldsymbol{x}^r\|^2] + \eta^2\sigma_g^2 \\ &\le (1-\mu\eta)\mathbb{E}[\|\boldsymbol{x}^* - \boldsymbol{x}^r\|^2] + \eta^2\sigma_g^2\end{aligned}$$

  where the last step holds if we choose η small enough

Now we see the same pattern as in the previous case. The proof can be immediately completed.

Convergence Analysis (f is Non-Convex)

Theorem 1.2 (Convergence of SGD for Non-Convex f)

Under the assumption in the previous page, where f is non-convex, if $\eta_r = \eta \leq \frac{1}{Lc_g}$, then SGD achieves the following rate

$$\frac{1}{T}\sum_{r=1}^{T}\mathbb{E}[\|\nabla F(\boldsymbol{x}^r)\|^2] \leq \frac{2(F(\boldsymbol{x}^0) - \underline{F})}{T\eta} + L\eta\sigma_g^2 \tag{1.7}$$

Proof Steps

- From the descent lemma, we have

$$
\begin{aligned}
F(\boldsymbol{x}^{r+1}) &\leq F(\boldsymbol{x}^r) + \langle \nabla F(\boldsymbol{x}^r), \boldsymbol{x}^{r+1} - \boldsymbol{x}^r \rangle + \frac{L}{2}\|\boldsymbol{x}^{r+1} - \boldsymbol{x}^r\|^2 \\
&= F(\boldsymbol{x}^r) - \eta \langle \nabla F(\boldsymbol{x}^r), \mathbf{g}^r \rangle + \frac{L\eta^2}{2}\|\mathbf{g}^r\|^2
\end{aligned}
$$

- Taking expectation, and use unbiasedness we obtain

$$
\begin{aligned}
F(\boldsymbol{x}^{r+1}) \leq F(\boldsymbol{x}^r) &- \eta \mathbb{E}[\langle \nabla F(\boldsymbol{x}^r), \mathbf{g}^r - \nabla F(x^r) \rangle] - \eta \|\nabla F(\boldsymbol{x}^r)\|^2 \\
&+ \frac{L\eta^2}{2}(\sigma_g^2 + c_g \|\nabla F(\boldsymbol{x})\|^2) \\
= F(\boldsymbol{x}^r) &- \left(\eta - \frac{L\eta^2 c_g}{2}\right)\|\nabla F(\boldsymbol{x}^r)\|^2 + \frac{L\eta^2}{2}\sigma_g^2
\end{aligned}
$$

Proof Steps

- Rearranging, and adding $r = 0, \cdots, T$

$$\left(\eta - \frac{L\eta^2 c_g}{2}\right) \frac{1}{T} \sum_{r=1}^{T} \|\nabla F(\boldsymbol{x}^r)\|^2 \leq \frac{F(\boldsymbol{x}^0) - F(\boldsymbol{x}^{T+1})}{T} + \frac{L\eta^2}{2}\sigma_g^2$$

- Let us pick η such that

$$\eta - \frac{L\eta^2 c_g}{2} \geq \eta/2, \text{ or } \eta \leq \frac{1}{Lc_g}$$

Then we obtain

$$\frac{1}{T} \sum_{r=1}^{T} \|\nabla F(\boldsymbol{x}^r)\|^2 \leq \frac{2(F(\boldsymbol{x}^0) - \underline{F})}{T\eta} + L\eta\sigma_g^2$$

Conclusion

- More references:

  1. Convergence of SGD under more settings

  2. Adam
  3. Pytorch implementation of Adam
  4. A summarization on different momentum-based optimizers